	Class	Adm No
Candidate Name: ______________________________		

Preliminary Examinations II 2009
Pre-university 3

H2 CHEMISTRY **9746 / 02**

Thursday 24 Sep 2009 1h 30 min

Additional materials:
Data Booklet

READ THESE INSTRUCTIONS FIRST

1. Do not turn over this question paper until you are told to do so.
2. Write your name, class and index number in the spaces provided at the top of this page.
3. Write in ***dark blue*** or ***black pen*** in the spaces provided on the Question Paper.
4. You may use a soft pencil for any diagrams or graphs.
5. **DO NOT** use paper clips, highlighters, glue or correction fluid or tape.
6. Answer **ALL** the questions.
7. Give non-exact numerical answers correct to ***3 significant figures***, or ***1 decimal place*** in the case of $\mathbf{M_r}$ and $\mathbf{A_r}$, unless a different level of accuracy is specified in the question.
8. The number of marks is given in brackets [ ] at the end of each question or part question.
9. You are reminded of the need for ***clear presentation*** in your answers and to ***show all working*** in calculations.
10. The use of a calculator is expected, where appropriate.

Paper 2							
Question No	**1**	**2**	**3**	**4**	**5**	**6**	
Marks Obtained							**60**

This question paper consists of 14 printed pages.

1. The half-reactions for a typical rechargeable nickel-cadmium ("nicad") battery are:

$$Cd(OH)_2(s) + 2e \rightleftharpoons Cd(s) + 2OH^-(aq) \qquad E^\theta = -0.81\ V$$

$$Ni(OH)_3(s) + e \rightleftharpoons Ni(OH)_2(s) + OH^-(aq) \qquad E^\theta = +0.49\ V$$

(a) Write a balanced equation that shows the reaction that occurs as the cell discharges. [1]

……………………………………………………………………………………………………...

(b) Using the information given above and the Data Booklet, can a nicad battery be used to drive the electrolytic decomposition of molten $MgCl_2$ into Mg metal and Cl_2 gas, i.e. $MgCl_2(l) \rightarrow Mg(s) + Cl_2(g)$? [3]

(c) If the nicad battery can deliver 0.1A for 12 hours, what is the mass of Mg can be produced from $MgCl_2$ by electrolysis? [2]

(d) The voltage delivered by a nicad battery does not change significantly as the battery discharges. Briefly explain why this is true. [1]

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

(e) Apart from the half equation given above, Ni also undergoes other redox processes.

$Ni^{2+} + 2e \rightleftharpoons Ni$ $\qquad E^{\theta} = -0.25\ V$

$[Ni(NH_3)_6]^{2+} + 2e \rightleftharpoons Ni + 6NH_3$ $\qquad E^{\theta} = -0.51\ V$

Suggest an explanation for the difference in the E^{θ} values observed. [2]

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

[Total: 9 marks]

2. An average eggshell weighs about 5 g. Most of the calcium in an eggshell is laid down within a 16h period. This means that it is deposited at a rate of about 125 mg per hour. The calcium required is supplied by special bony masses in the hen's long bones, which accumulate large reserves of calcium for eggshell formation. [The inorganic calcium component of the bone is calcium phosphate, $Ca_3(PO_4)_2$, an insoluble compound.]

   The eggshell is largely composed of calcite, a crystalline form of calcium carbonate ($CaCO_3$). Normally, the raw materials, Ca^{2+} and CO_3^{2-}, are carried by the blood to the shell gland. The calcification process is as shown:

$$Ca^{2+}(aq) + CO_3^{2-}(aq) \rightleftharpoons CaCO_3(s)$$

   In the blood, free Ca^{2+} ions are in equilibrium with calcium ions bound to proteins. As the free ions are taken up by the shell gland, more are provided by the dissociation of the protein-bound calcium.

   The carbonate ions necessary for eggshell formation are a metabolic byproduct. Carbon dioxide produced during metabolism is converted to carbonic acid (H_2CO_3) by the enzyme carbonic anhydrase (CA):

$$CO_2(g) + H_2O(l) \xrightleftharpoons{CA} H_2CO_3(aq)$$

   Carbonic acid ionizes stepwise to produce carbonate ions:

$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$
$$HCO_3^-(aq) \rightleftharpoons H^+(aq) + CO_3^{2-}(aq)$$

   The carbonic acid formed helps to set up a H_2CO_3/HCO_3^- buffer system in the blood plasma of the hen where the pH is maintained at about 7.4. Variation in pH of the blood plasma may alter the structure of important proteins and hence affect certain biological processes.

   Chickens do not perspire and so must pant to cool themselves. Panting expels more CO_2 from the chicken's body than normal respiration does.

(a) (i) Calculate the mass of calcium phosphate required to form one egg. [2]

(ii) Suggest whether eggs produced by the hen on a hot day have thicker or thinner shells when compared to those produced on a cooler day. Explain your answer. [2]

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

(iii) Suggest one remedy to compensate for the lost of CO_2 from the hen on a hot day. [1]

…………………………………………………………………………………………………

(iv) Write equations to show how the blood plasma of the hen acts as a buffer. [2]

…………………………………………………………………………………………………

…………………………………………………………………………………………………

(v) With reference to a biological function of proteins, explain how variation in pH alters the structure of proteins and hence affect its function. [3]

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

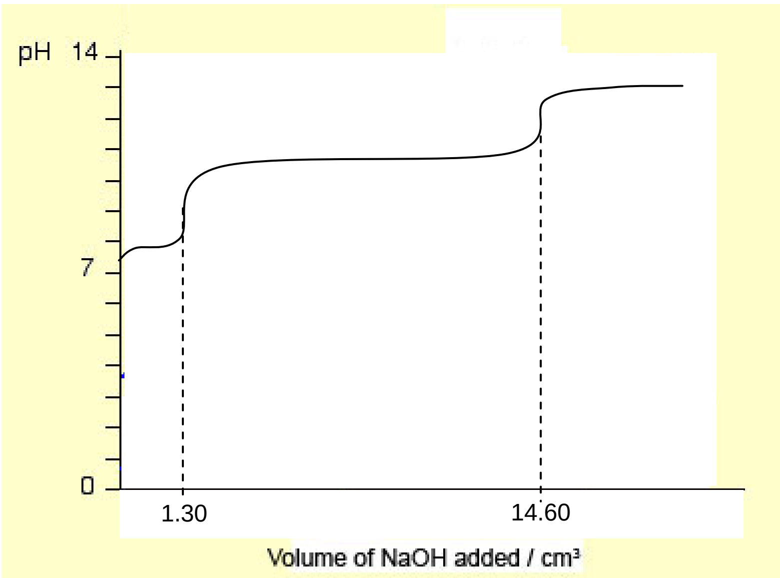
…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

(b) A 10 cm^3 sample of blood plasma was drawn from the hen and titrated against a solution of 7.50×10^{-3} mol dm^{-3} NaOH, using two different indicators, to determine the amount of each of the substances that make up the buffer. A graph of pH against volume of NaOH was obtained.



(i) Calculate the amount of H_2CO_3 present in the sample of blood plasma. [1]

(ii) Calculate the amount of HCO_3^- present in the sample of blood plasma. [2]

(iii) Hence determine the K_a of carbonic acid. [2]

(iv) The theoretical value for the K_a of carbonic acid is 4.47×10^{-7} mol dm^{-3}. Suggest an error or limitation of this experiment which would have resulted in the difference in the value calculated in (b)(iii). [1]

……………………………………………………………………………………………

……………………………………………………………………………………………

(v) Using the theoretical value for the K_a of carbonic acid, calculate the pH of the first endpoint. [4]

[Total: 20 marks]

3. (a) Oxy salts such as carbonates, nitrates and sulphates can thermally decompose.

Describe and explain how the decomposition temperature of barium carbonate compares with that of strontium carbonate. [3]

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

(b) Magnesium nitrate, $Mg(NO_3)_2$, thermally decomposes to form magnesium oxide, oxygen and nitrogen dioxide.

(i) Write the equation for this decomposition. [1]

……………………………………………………………………………………………

(ii) Explain why the lattice energy of magnesium oxide is much more exothermic than that of magnesium nitrate. [2]

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

(c) MgO, Al_2O_3 and SiO_2 are all white solids. Suggest how they may be differentiated from one another. Write balanced equations for any reactions that occur. [3]

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

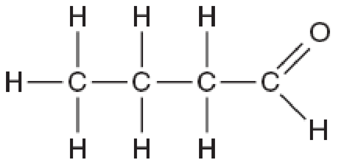
……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

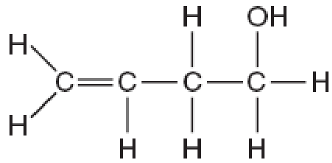
……………………………………………………………………………………………………..

[Total: 9 marks]

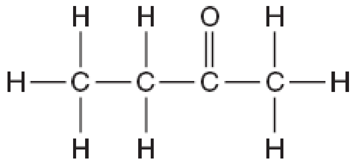
4. An unknown colourless liquid with molecular formula C_4H_8O was thought to be one of butanal, but-3-en-1-ol, or butanone.



butanal



but-3-en-1-ol



butanone

(a) State a simple chemical test that would identify:

(i) butanal only;

Reagent:	
Condition:	
Observation:	
Balanced equation:	

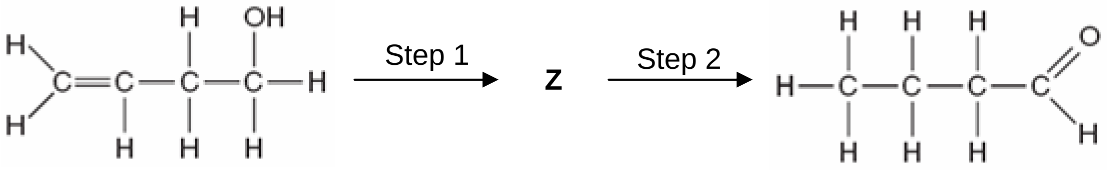
(ii) but–3–en–1–ol only. [6]

Reagent:	
Condition:	
Observation:	
Balanced equation:	

(b) Draw the product formed when butanone reacts with 2,4 – dinitrophenyhydrazine.

[1]

(c) But–3–en–1–ol can be converted to butanal in the following reaction scheme.



(i) Draw the displayed structure of **Z**. [1]

(ii) State the reagents and conditions required for both steps. [2]

	Reagents	Conditions
Step 1		
Step 2		

[Total: 10 marks]

5. (a) Nitration of benzene using concentrated nitric acid and a suitable catalyst is an important industrial process.

(i) Outline the mechanism for the nitration of benzene, starting from nitric acid, to form nitrobenzene. Your answer should also show how the catalyst is involved. [4]

(ii) This preparation of nitrobenzene is generally carried out at 55 – 60°C. If the temperature rises above 100°C, a compound forms with the formula $C_6H_4N_2O_4$. Suggest the structure of this compound. [1]

(b) Benzene also reacts with bromine via the same mechanism, resulting in the decolourisation of the brown liquid. Decolourisation of bromine occurs more rapidly when reacted with phenol.

Explain the different reactivities of bromine with benzene and with phenol.

[2]

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

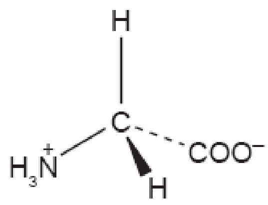
………………………………………………………………………………………………

[Total: 7 marks]

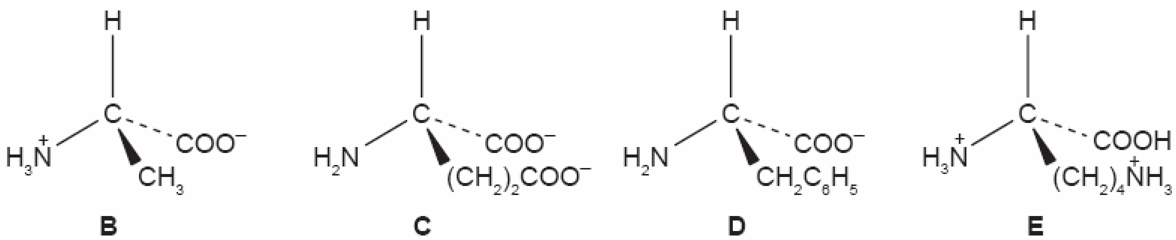
6. Amino acids are the fundamental units to building a protein. The amino acids present in a polypeptide chain greatly affect the type of interactions present. This in turn affects the shape of the protein and also the specificity of its functions.

(a) Amino acids can be separated by electrophoresis. The separation depends on various factors. One of which is the pH of the mixture.

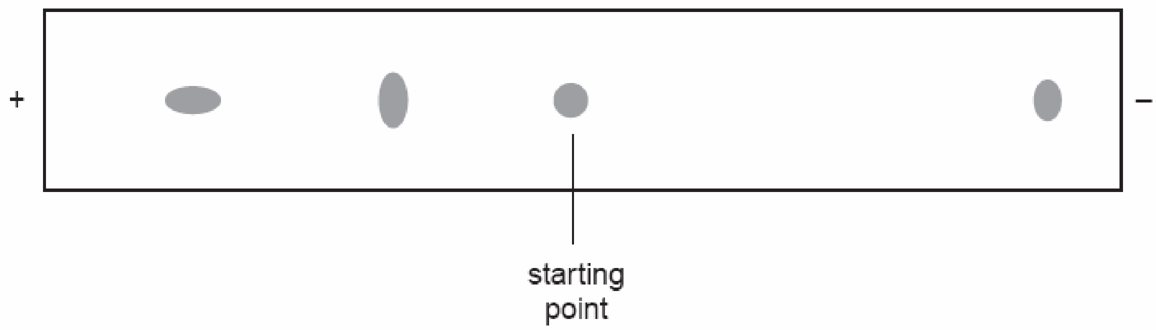
At a pH of 5.8, glycine forms a zwitterion.



At a pH of 6.0, the four amino acids in a mixture exist in the forms **B – E** shown below.

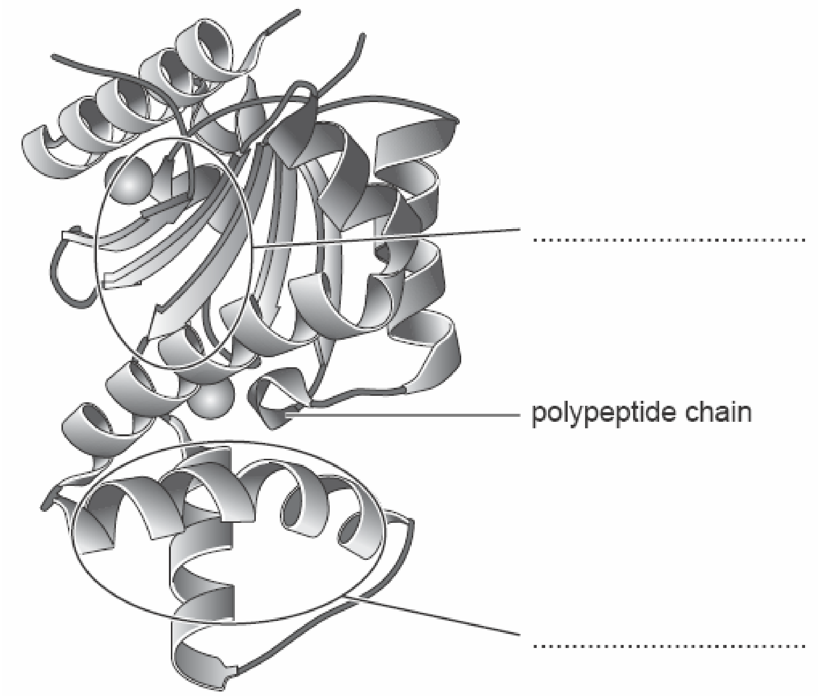


The diagram below shows the result of the electrophoresis of the mixture of the four amino acids at pH 6.0.



Label the spots **B**, **C**, **D** and **E** appropriately. [2]

(b) The diagram represents the structure of the bacterial enzyme cellulase, showing its secondary and tertiary structure.



(i) On the diagram above, complete the labels for each type of secondary structure. [1]

(ii) Explain what holds each type of secondary structure together. Your answer should include a diagram, with any atoms of groups involved clearly shown. [2]

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

[Total: 5 marks]

-END OF PAPER-